

Hector V3.2.0 Tutorial

Hector is an open-source, object-oriented, Reduced Complexity Earth Systems Model (RCM). It models critical earth system processes on a global scale. As a computationally efficient model, Hector has many applications. It can be used to explore scenarios, generate large ensembles of results, explore key uncertainties, and emulate Earth System Models. There are three ways to use Hector: as a stand-alone executable, through the online HectorUI, and as an R package. Installation instructions and examples are available at <https://github.com/jgcri/hector> and <https://github.com/jgcri/hectorui>.

Follow the appropriate [installation for your machine](#) to run locally. Here you will learn how to use the R hector package to run complete a Hector run, change a model parameter, and query results.

Getting started

```
In [1]: # Load the Hector package (this requires that Hector has already been installed
# see http://jgcri.github.io/hector/articles/BuildHector.html)
library(hector)

# Package used to plot results
library(ggplot2)
theme_set(theme_bw(base_size = 20)) # plot setting
options(repr.plot.width=16, repr.plot.height=8)
```

The Hector R package ships with a handful of scenarios, located in the input directory of the package. See a list of the ini (initialization) files using the `system.file` function. Each one of these files is a configuration file for a hector run, each file contains the default model parameters and the emission time-series for the SSP scenarios.

```
In [2]: files <- list.files(system.file("input", package = "hector"))
cat(files, sep= "\n")
```

```
hector_ssp119.ini
hector_ssp126.ini
hector_ssp245.ini
hector_ssp370.ini
hector_ssp434.ini
hector_ssp460.ini
hector_ssp534-over.ini
hector_ssp585.ini
tables
```

Run SSP2-4.5

Use the `newcore` function to set up a Hector core to run the SSP2-4.5 scenario.

```
In [3]: # Select the ini file to use.
ini_file = system.file("input/hector_ssp245.ini", package = "hector")

# Set up a Hector core using the ini file.
hcore = newcore(inifile = ini_file, name = "SSP2-45")

# Run Hector! It will take less than a second.
run(core = hcore)
```

```
Hector core:    SSP2-45
Start date:    1745
End date:      2300
Current date:  2300
Input file:    C:\Users\brow521\AppData\Local\R\win-library\4.4\hector\input\hector
_ssp245.ini
```

Use the `fetchvars` function to query the Hector results. It can be used to get a time series of results or model parameters.

```
In [4]: # A vector of the variables we are interested in. Hector has a number of helper fun
# string names recognized by the Hector core.
vars = c(GLOBAL_TAS(), CONCENTRATIONS_CO2())
cat(vars, sep = "\n")
```

```
global_tas
CO2_concentration
```

```
In [5]: out_ssp245 = fetchvars(core = hcore,          # the active hector core
                               dates = 1850:2100,    # the vector contains the years of int
                               vars = vars)           # the vector contains the variables of

head(out_ssp245)
```

A data.frame: 6 × 5

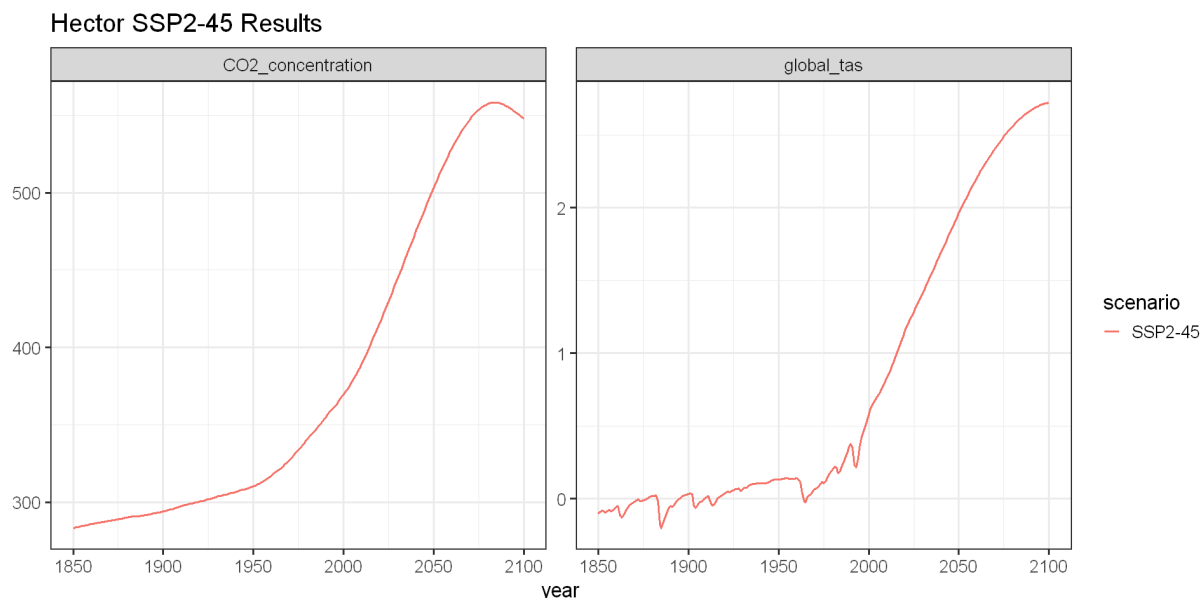
	scenario	year	variable	value	units
	<chr>	<dbl>	<chr>	<dbl>	<chr>
1	SSP2-45	1850	global_tas	-0.10249176	degC
2	SSP2-45	1851	global_tas	-0.09068423	degC
3	SSP2-45	1852	global_tas	-0.08036054	degC
4	SSP2-45	1853	global_tas	-0.08557470	degC
5	SSP2-45	1854	global_tas	-0.09308445	degC
6	SSP2-45	1855	global_tas	-0.08554412	degC

```
In [6]: # After fetching the results from Hector shut down the Hector core, this is a good
shutdown(hcore)
```

Hector core (INACTIVE)

Now we can plot the Hector results.

```
In [7]: ggplot(data = out_ssp245) +
  geom_line(aes(year, value, color = scenario), linewidth = 1) +
  labs(title = "Hector SSP2-45 Results") +
  facet_wrap("variable", scales = "free") +
  labs(y = NULL)
```



How to change a Hector parameter

In this example we will learn how to change Hector's equilibrium climate sensitivity parameter (`ECS`).

```
In [8]: # First set up a Hector core
ini_file = system.file("input/hector_ssp245.ini", package = "hector")
hcore = newcore(inifile = ini_file, name = "SSP2-45 higher ECS")
run(core = hcore)
```

```
Hector core:    SSP2-45 higher ECS
Start date:     1745
End date:       2300
Current date:   2300
Input file:     C:\Users\brow521\AppData\Local\R\win-library\4.4\hector\input\hector_ssp245.ini
```

Use the `fetchvar` to extract the default ECS value.

```
In [9]: default_ECS = fetchvars(core = hcore, dates = NA, vars = ECS())
default_ECS
```

A data.frame: 1 × 5

scenario	year	variable	value	units
<chr>	<dbl>	<chr>	<dbl>	<chr>
SSP2-45 higher ECS	NA	S	3	degC

Here, we will double the default ECS and pass the information to the Hector core by calling `setvar()`. The `setvar()` function needs the string name for units, which we can get from the helper `getunits()` function.

```
In [10]: # double the default ECS value
double_ECS = 2 * default_ECS$value
double_ECS
```

6

```
In [11]: ECS_units = getunits(vars = ECS())
ECS_units
```

'degC'

```
In [12]: setvar(core = hcore,
              dates = NA,
              var = ECS(),
              values = double_ECS,
              unit = ECS_units)
```

```
In [13]: run(core = hcore)
```

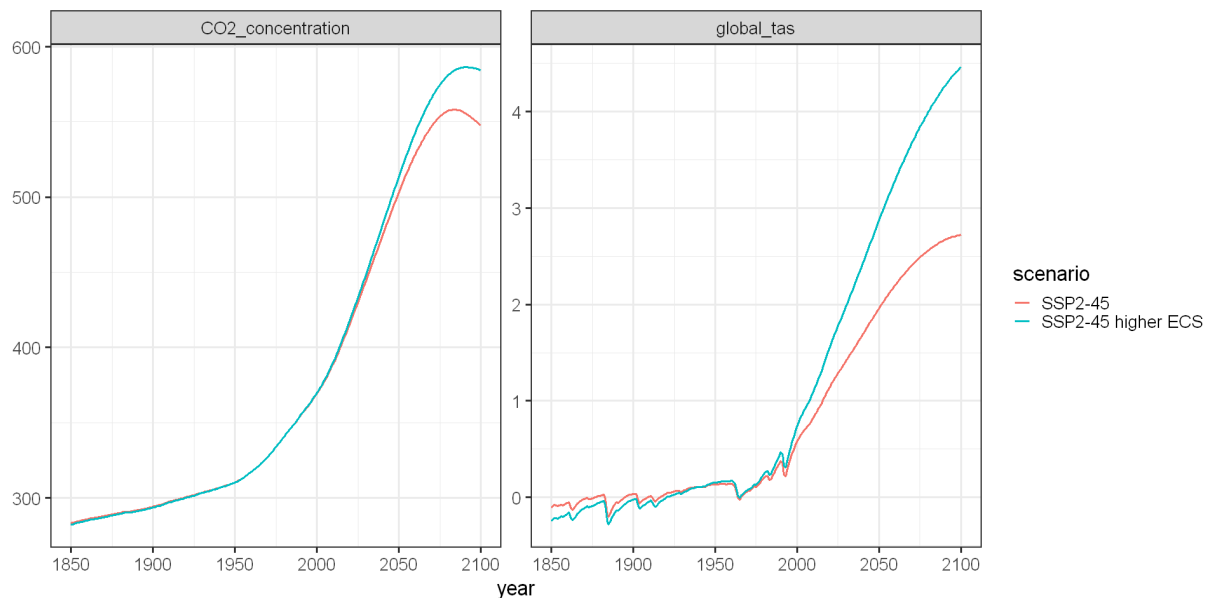
Auto-resetting core to 0

```
Hector core:    SSP2-45 higher ECS
Start date:     1745
End date:       2300
Current date:   2300
Input file:     C:\Users\brow521\AppData\Local\R\win-library\4.4\hector\input\hector
_ssp245.ini
```

```
In [14]: out_ssp245_newECS = fetchvars(core = hcore, dates = 1850:2100, vars = vars)
```

```
In [15]: # Combine the data frames into a single data frame, then plot it.
results <- rbind(out_ssp245, out_ssp245_newECS)

ggplot(data = results) +
  geom_line(aes(year, value, color = scenario), linewidth = 1) +
  facet_wrap("variable", scales = "free") +
  labs(y = NULL)
```



Modifying an emissions pathway

RCMS such as Hector can be used to explore custom/new emissions scenarios. Using the `setvars` function users can modify the emission input pathways, allowing users to test out new scenarios. Here, we will modify the CO₂ emissions from fossil fuel and industry (`ffi_emissions`) to demonstrate how to change emission pathways. **Note** that a total of 40 different emission pathways can be set using this approach, `ffi_emissions` is just the example we are using here.

```
In [16]: # Run to see documentation on Hector emissions
         help(emissions)
```

Identifiers for miscellaneous emissions not elsewhere described

Description

All of these variables can be set using the `SETDATA` message type.

Usage

`EMISSIONS_BC()`

`EMISSIONS_N2O()`

`NAT_EMISSIONS_N2O()`

`EMISSIONS_NOX()`

`EMISSIONS_CO()`

`EMISSIONS_NMVOC()`

`EMISSIONS_OC()`

`EMISSIONS_NH3()`

Functions

- `EMISSIONS_BC()` : Black carbon emissions
- `EMISSIONS_N2O()` : N2O emissions ("Tg N")
- `NAT_EMISSIONS_N2O()` : Natural N2O emissions ("Tg N")
- `EMISSIONS_NOX()` : Emissions for NOx compounds ("Tg N")
- `EMISSIONS_CO()` : Emissions for carbon monoxide ("Tg CO")
- `EMISSIONS_NMVOC()` : Emissions for non-methane volatile organic compounds (NMVOC) ("Tg NMVOC")
- `EMISSIONS_OC()` : Emissions for organic carbon ("Tg")
- `EMISSIONS_NH3()` : Emissions for NH3 ("Tg")

Note

Because these identifiers are provided as `#define` macros in the hector code, these identifiers are provided in the R interface as functions. Therefore, these objects must be called to use them; e.g., `GETDATA()` instead of the more natural looking `GETDATA`.

See Also

Other capability identifiers: `CF4_CONSTRAIN()`, `CONCENTRATIONS_CH4()`, `CONCENTRATIONS_N2O()`, `DELTA_CO2()`, `EMISSIONS_CF4()`, `EMISSIONS_SO2()`, `FTOT_CONSTRAIN()`, `GLOBAL_TAS()`, `NBP()`, `OCEAN_UPTAKE()`, `RF_CF4()`, `RF_TOTAL()`, `RHO_BC()`, `TRACKING_DATE()`

[Package *hector* version 3.2.0]

In [17]: *# Again start off with an active Hector core and complete a default run so that we*

```
ini_file = system.file("input/hector_ssp245.ini", package = "hector")
hcore = newcore(inifile = ini_file, name = "SSP2-45")
run(core = hcore)
```

```
Hector core:    SSP2-45
Start date:    1745
End date:      2300
Current date:  2300
Input file:    C:\Users\brow521\AppData\Local\R\win-library\4.4\hector\input\hector
               _ssp245.ini
```

In [18]: *# Fetch a copy of the global temperature and FFI emissions to compare with new run.*

```
original_results <- fetchvars(core = hcore,
                              dates = 1850:2100,                # date(s) i
                              vars = c(FFI_EMISSIONS(), GLOBAL_TAS())) # variable
original_results$scenario <- "Original SSP2-45"
```

In [19]: *# Save a copy the FFI emissions pathway, this is the dataframe we will be manipulat*

```
ffi_emissions <- fetchvars(core = hcore,
                           dates = 1850:2100,                # date(s) interested in
                           vars = FFI_EMISSIONS())           # variable(s) intrested in

head(ffi_emissions)
```

A data.frame: 6 × 5

	scenario	year	variable	value	units
	<chr>	<dbl>	<chr>	<dbl>	<chr>
1	SSP2-45	1850	ffi_emissions	0.05138490	Pg C/yr
2	SSP2-45	1851	ffi_emissions	0.05156231	Pg C/yr
3	SSP2-45	1852	ffi_emissions	0.05364238	Pg C/yr
4	SSP2-45	1853	ffi_emissions	0.05595235	Pg C/yr
5	SSP2-45	1854	ffi_emissions	0.06589565	Pg C/yr
6	SSP2-45	1855	ffi_emissions	0.06656059	Pg C/yr

Now, for the sake of this example let's cut the emissions in half.

```
In [20]: ffi_emissions$value <- ffi_emissions$value * 0.5
```

```
In [21]: head(ffi_emissions)
```

A data.frame: 6 × 5

	scenario	year	variable	value	units
	<chr>	<dbl>	<chr>	<dbl>	<chr>
1	SSP2-45	1850	ffi_emissions	0.02569245	Pg C/yr
2	SSP2-45	1851	ffi_emissions	0.02578115	Pg C/yr
3	SSP2-45	1852	ffi_emissions	0.02682119	Pg C/yr
4	SSP2-45	1853	ffi_emissions	0.02797617	Pg C/yr
5	SSP2-45	1854	ffi_emissions	0.03294783	Pg C/yr
6	SSP2-45	1855	ffi_emissions	0.03328030	Pg C/yr

```
In [22]: # Use setvars to pass in the new emissions pathway.
setvar(core = hcore,
       dates = ffi_emissions$year,
       var = ffi_emissions$variable,
       values = ffi_emissions$value,
       unit = ffi_emissions$units)
run(hcore)
```

Auto-resetting core to 1849

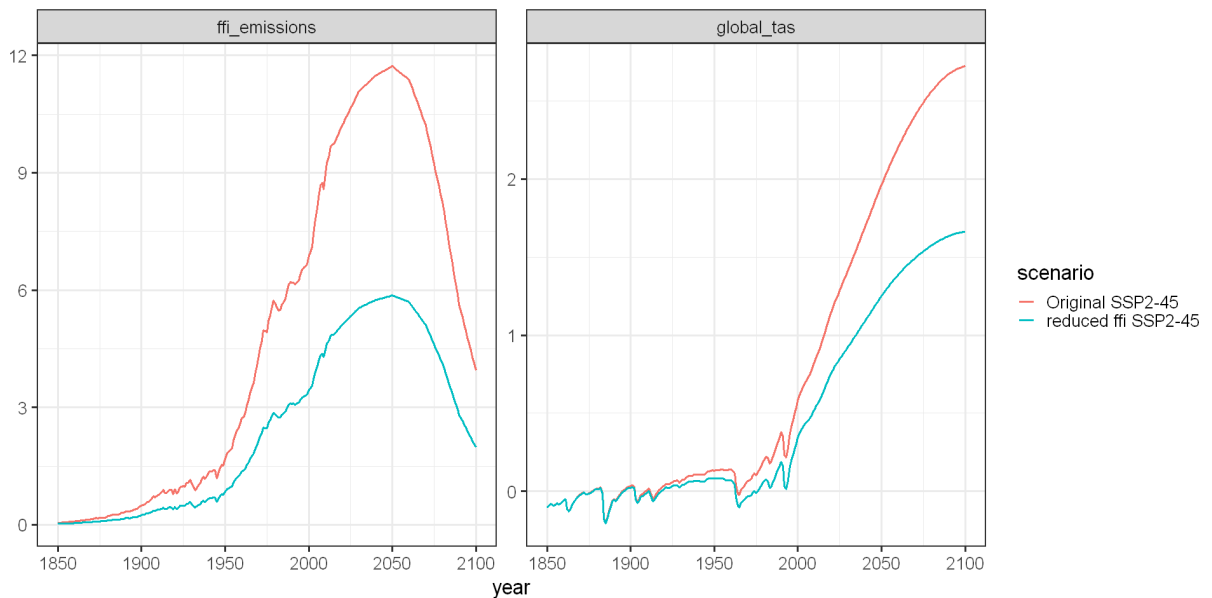
```
Hector core:    SSP2-45
Start date:     1745
End date:       2300
Current date:   2300
Input file:     C:\Users\brow521\AppData\Local\R\win-library\4.4\hector\input\hector
                _ssp245.ini
```



```
In [23]: # Query results
reduced_ffl_results <- fetchvars(core = hcore,
                                dates = 1850:2100,                                # date(s) i
                                vars = c(FFI_EMISSIONS(), GLOBAL_TAS()))            # variable
reduced_ffl_results$scenario <- "reduced ffl SSP2-45"
```

```
In [24]: # Combine the data frames into a single data frame, then plot it.
results <- rbind(original_results, reduced_ffl_results)

ggplot(data = results) +
  geom_line(aes(year, value, color = scenario), linewidth = 1) +
  facet_wrap("variable", scales = "free") +
  labs(y = NULL)
```



Check out more examples at <http://jgcri.github.io/hector/>.